

# Software Requirements Specification

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## **\*\*Software Requirements Specification (SRS)\*\***

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### **\*\*Introduction\*\***

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This Software Requirements Specification (SRS) document outlines the requirements for the development of the B09 application. The application is designed to provide a user-friendly interface for managing activities, badges, and user interactions. The SRS document will serve as a guide for the development team to ensure that the application meets the required functionality, performance, and quality standards.

### **\*\*Purpose\*\***

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The purpose of this SRS document is to:

- \* Define the requirements for the B09 application
- \* Provide a clear understanding of the application's functionality and interfaces
- \* Ensure that the development team has a common understanding of the application's requirements
- \* Serve as a basis for testing and validation of the application

### **\*\*Scope\*\***

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The scope of this SRS document includes the following components:

- \* Backend API (server-side)
- \* Frontend API (client-side)
- \* Mobile application
- \* Database models (Activity, Badge, User, Image)

The scope excludes the following components:

- \* Third-party services (e.g., authentication, payment gateways)
- \* External APIs (e.g., social media, mapping services)

### **\*\*System Overview\*\***

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The B09 application is a web-based and mobile application that allows users to manage activities, badges, and user interactions. The application consists of the following components:

- \* Backend API: responsible for handling API calls, database interactions, and business logic
- \* Frontend API: responsible for handling client-side logic, rendering, and user interactions
- \* Mobile application: provides a native mobile experience for users
- \* Database models: define the structure and relationships between data entities (Activity, Badge, User, Image)

#### **\*\*Functional Requirements\*\***

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The following are the functional requirements for the B09 application:

##### 1. **\*\*User Management\*\***:

- \* Register new users (POST /register)
- \* Complete levels (POST /complete-level)

##### 2. **\*\*Activity Management\*\***:

- \* Create new activities (POST /)
- \* Get activities (GET /)
- \* Complete activities (POST /user)

##### 3. **\*\*Badge Management\*\***:

- \* Create new badges (POST /)
- \* Get badges (GET /)

##### 4. **\*\*Activity Routes\*\***:

- \* Get activities (GET /)
- \* Create new activities (POST /)

##### 5. **\*\*Summarize Routes\*\***:

- \* Get summarize data (POST /)
- \* Ask eco questions (POST /ask-eco)

##### 6. **\*\*Leaderboard Routes\*\***:

\* Get leaderboard data (GET /)

## **\*\*Non-Functional Requirements\*\***

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The following are the non-functional requirements for the B09 application:

1. **\*\*Performance\*\***: The application should respond to user interactions within 2 seconds.
2. **\*\*Security\*\***: The application should implement authentication and authorization mechanisms to protect user data.
3. **\*\*Usability\*\***: The application should provide an intuitive and user-friendly interface.
4. **\*\*Scalability\*\***: The application should be able to handle increased traffic and user growth.

## **\*\*Architecture\*\***

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The B09 application will use the following architecture:

- \* Backend API: built using Node.js, Express.js, and MongoDB
- \* Frontend API: built using React.js and APIs
- \* Mobile application: built using React Native
- \* Database models: defined using MongoDB and Mongoose

## **\*\*Risks\*\***

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The following are potential risks associated with the B09 application:

1. **\*\*Technical debt\*\***: The application may incur technical debt due to the use of outdated technologies or frameworks.
2. **\*\*Security vulnerabilities\*\***: The application may be vulnerable to security threats if proper authentication and authorization mechanisms are not implemented.
3. **\*\*Scalability issues\*\***: The application may experience scalability issues if not designed to handle increased traffic and user growth.

## **\*\*Constraints\*\***

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The following are constraints that may impact the development of the B09 application:

1. **\*\*Time constraints\*\***: The application must be developed within a tight timeline.

2. **Resource constraints**: The development team may have limited resources (e.g., personnel, budget).

3. **Technical constraints**: The application may be limited by technical constraints (e.g., outdated technologies, framework limitations).

#### **Future Scope**

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The following are potential future scope items for the B09 application:

1. **Integration with third-party services**: The application may be integrated with third-party services (e.g., social media, payment gateways).

2. **Additional features**: The application may include additional features (e.g., gamification, personalized recommendations).

3. **Expansion to new platforms**: The application may be expanded to new platforms (e.g., desktop, TV).

By following this SRS document, the development team can ensure that the B09 application meets the required functionality, performance, and quality standards, and provides a user-friendly interface for managing activities, badges, and user interactions.